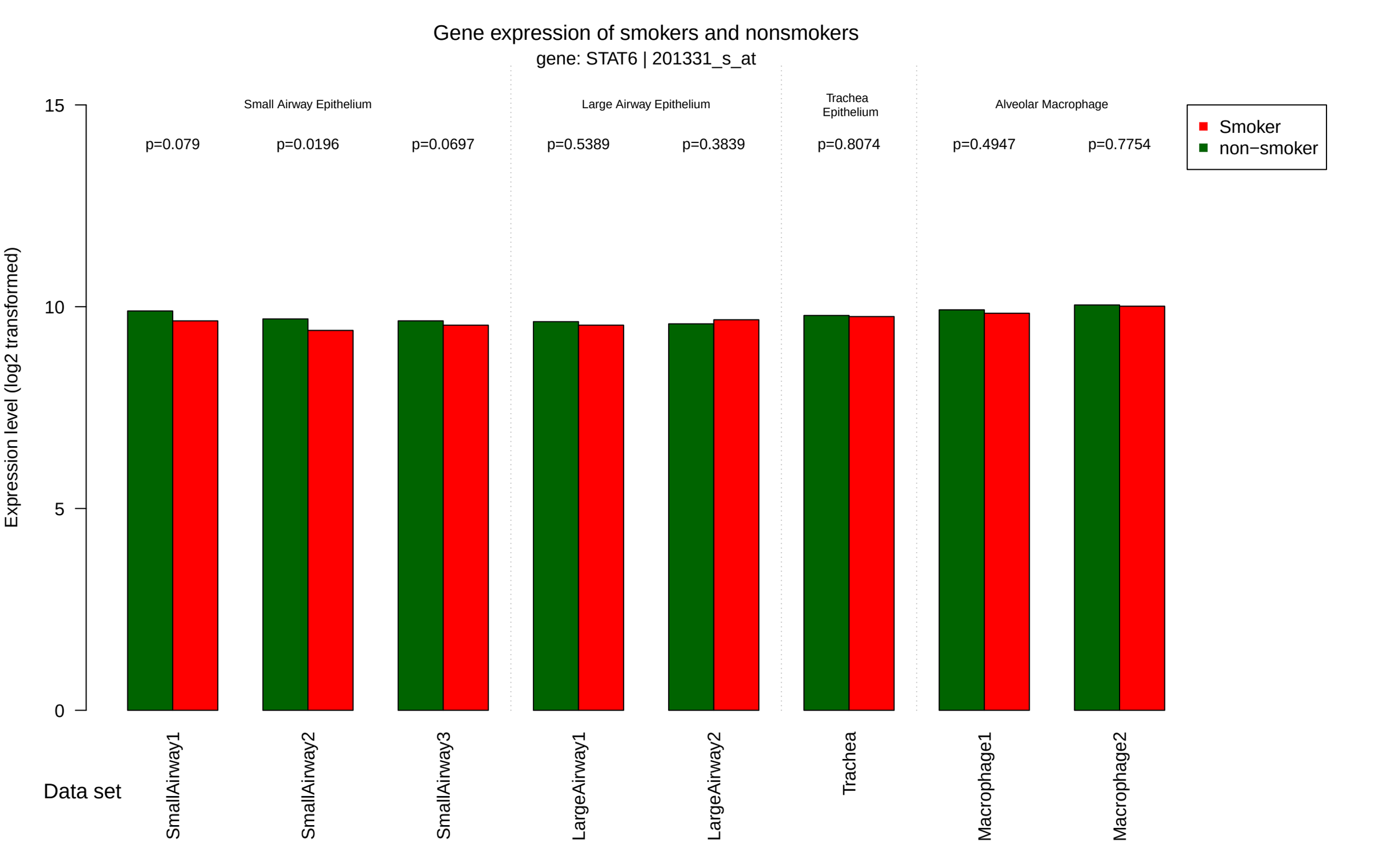
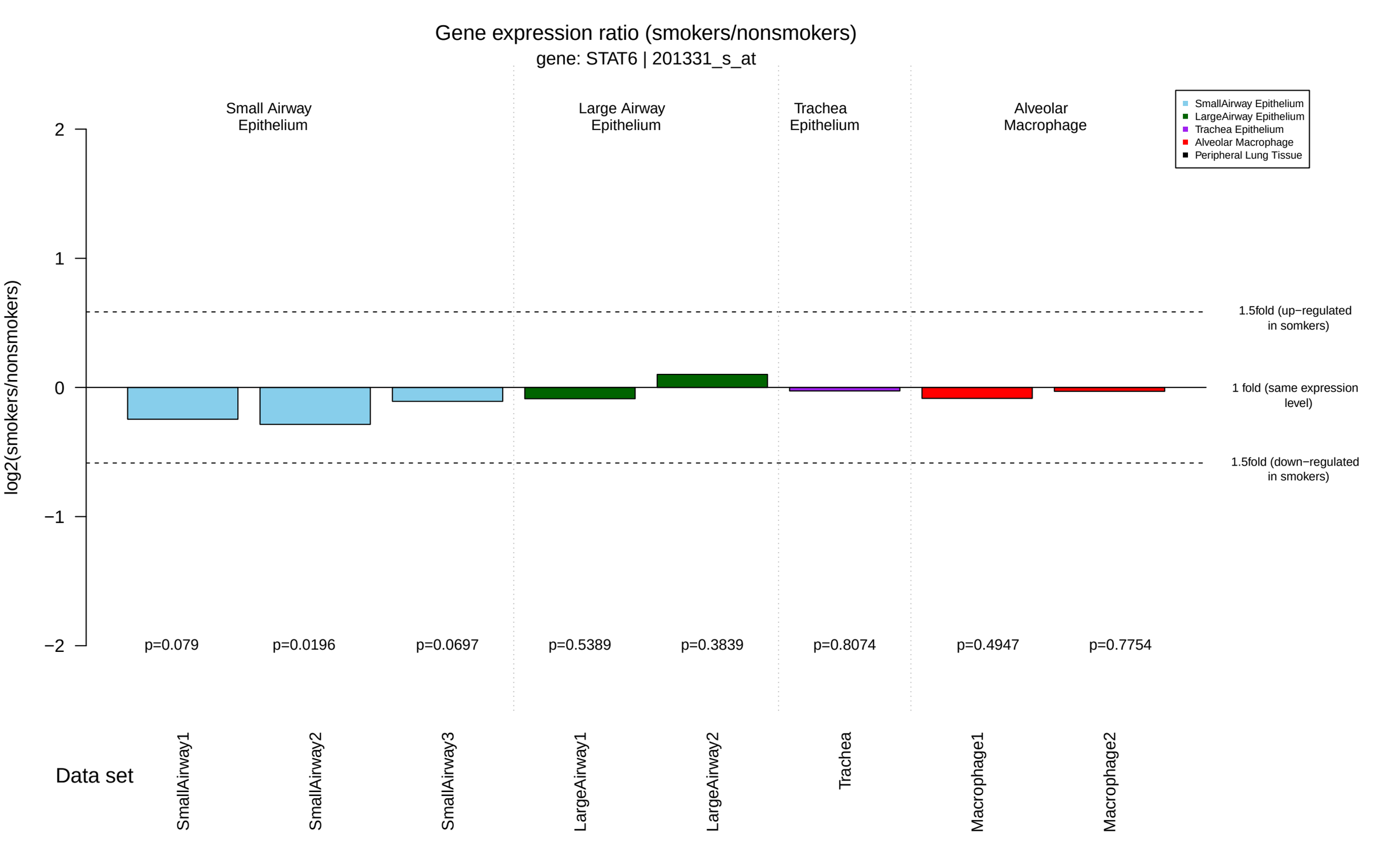
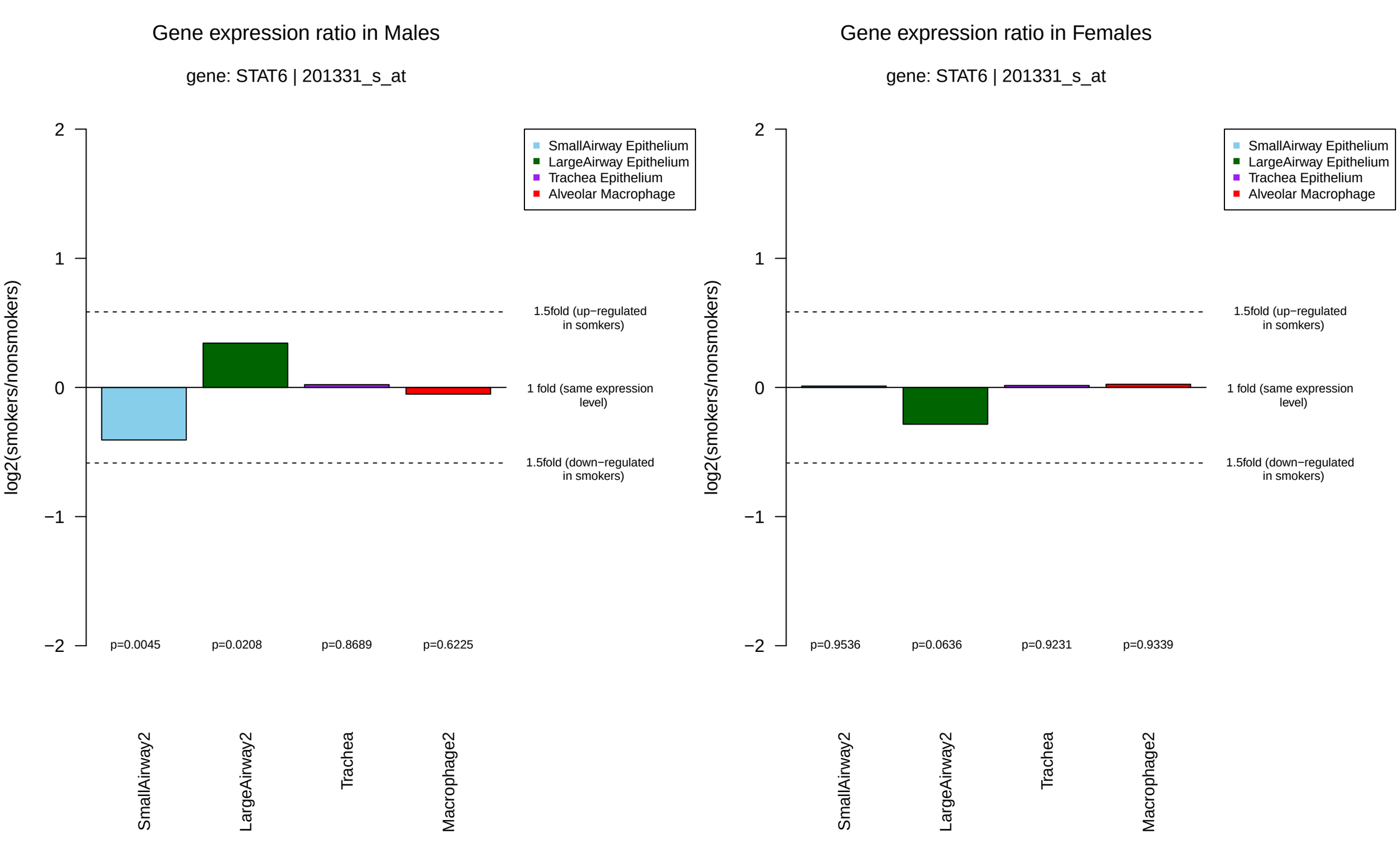
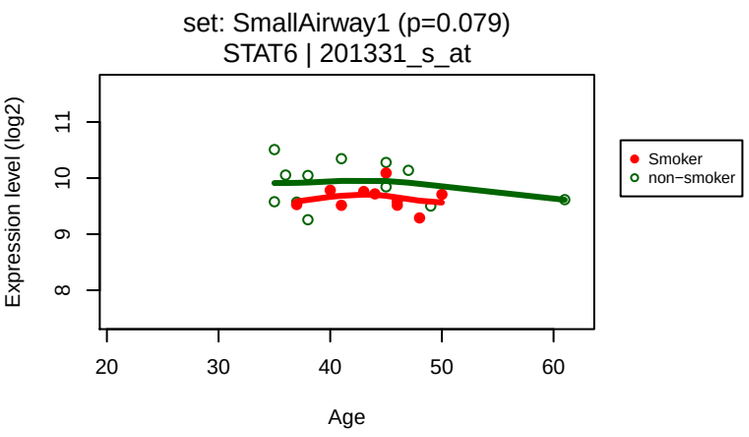
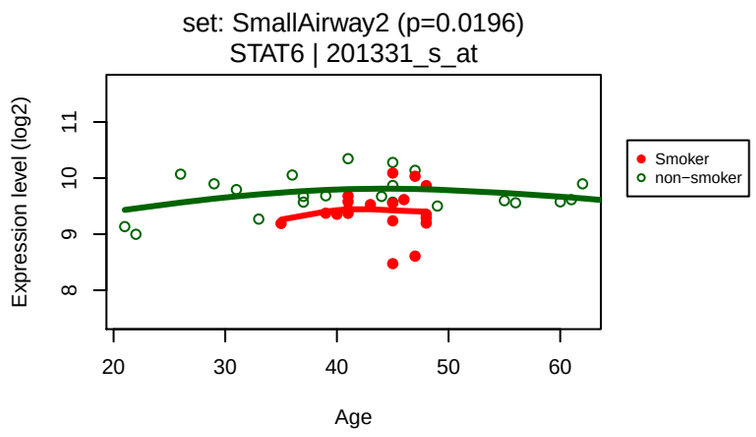
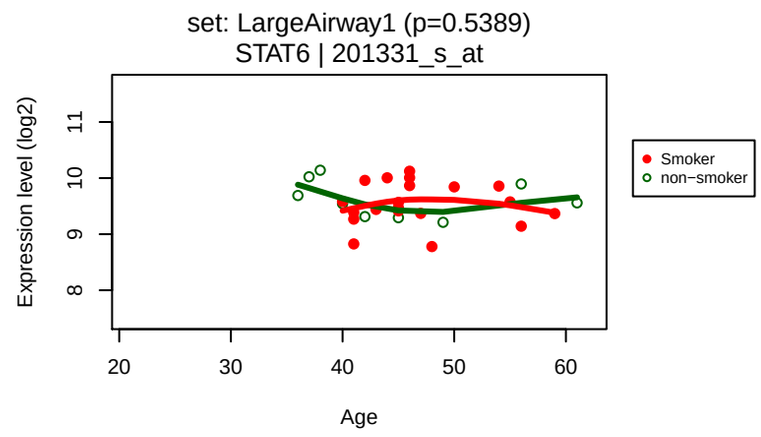

Gene expression of smokers and nonsmokers
gene: STAT6 | 201331_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.079
p=0.0196
p=0.0697
p=0.5389
p=0.3839
p=0.8074
p=0.4947
p=0.7754
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

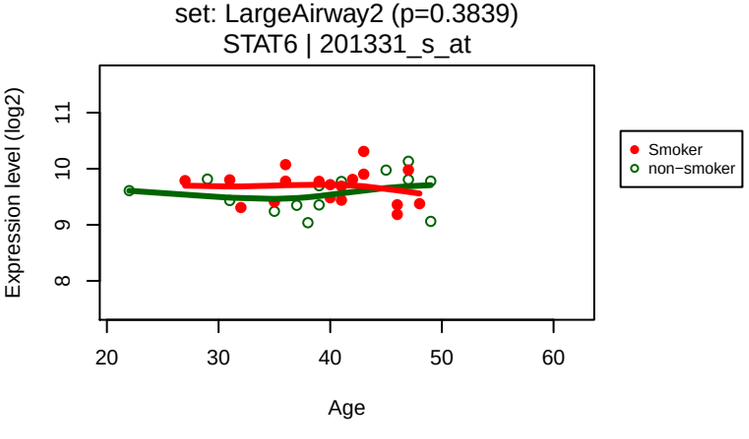

Gene expression ratio (smokers/nonsmokers)
gene: STAT6 | 201331_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.079
p=0.0196
p=0.0697
p=0.5389
p=0.3839
p=0.8074
p=0.4947
p=0.7754
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

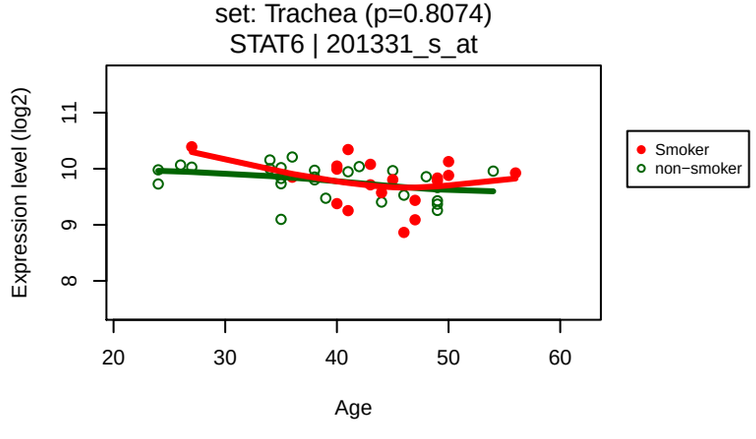

Gene expression ratio in Males
gene: STAT6 | 201331_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0045
p=0.0208
p=0.8689
p=0.6225
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT6 | 201331_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9536
p=0.0636
p=0.9231
p=0.9339
SmallAirway2
LargeAirway2
Trachea
Macrophage2

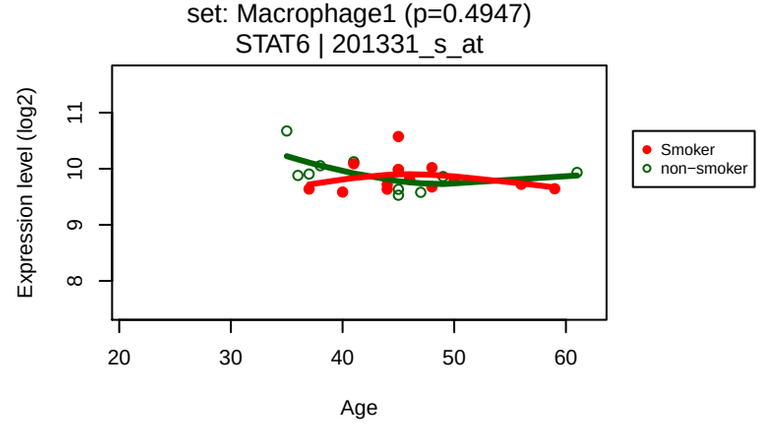

set: SmallAirway1 (p=0.079)
STAT6 | 201331_s_at
Expression level (log2)
Age
Smoker
non−smoker

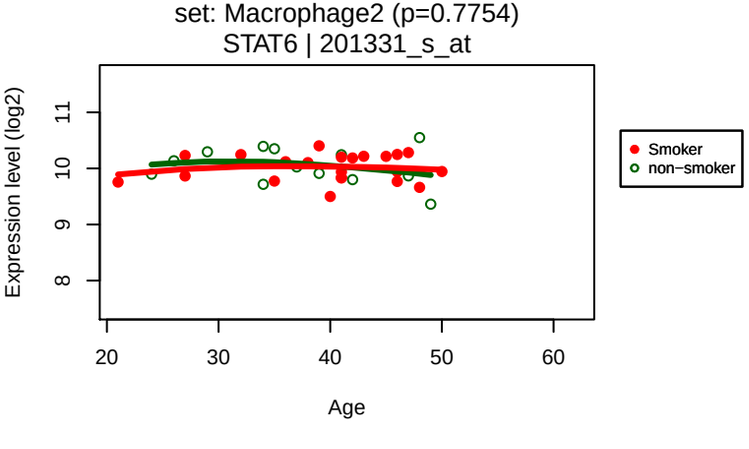

set: SmallAirway2 (p=0.0196)
STAT6 | 201331_s_at
Expression level (log2)
Age
Smoker
non−smoker

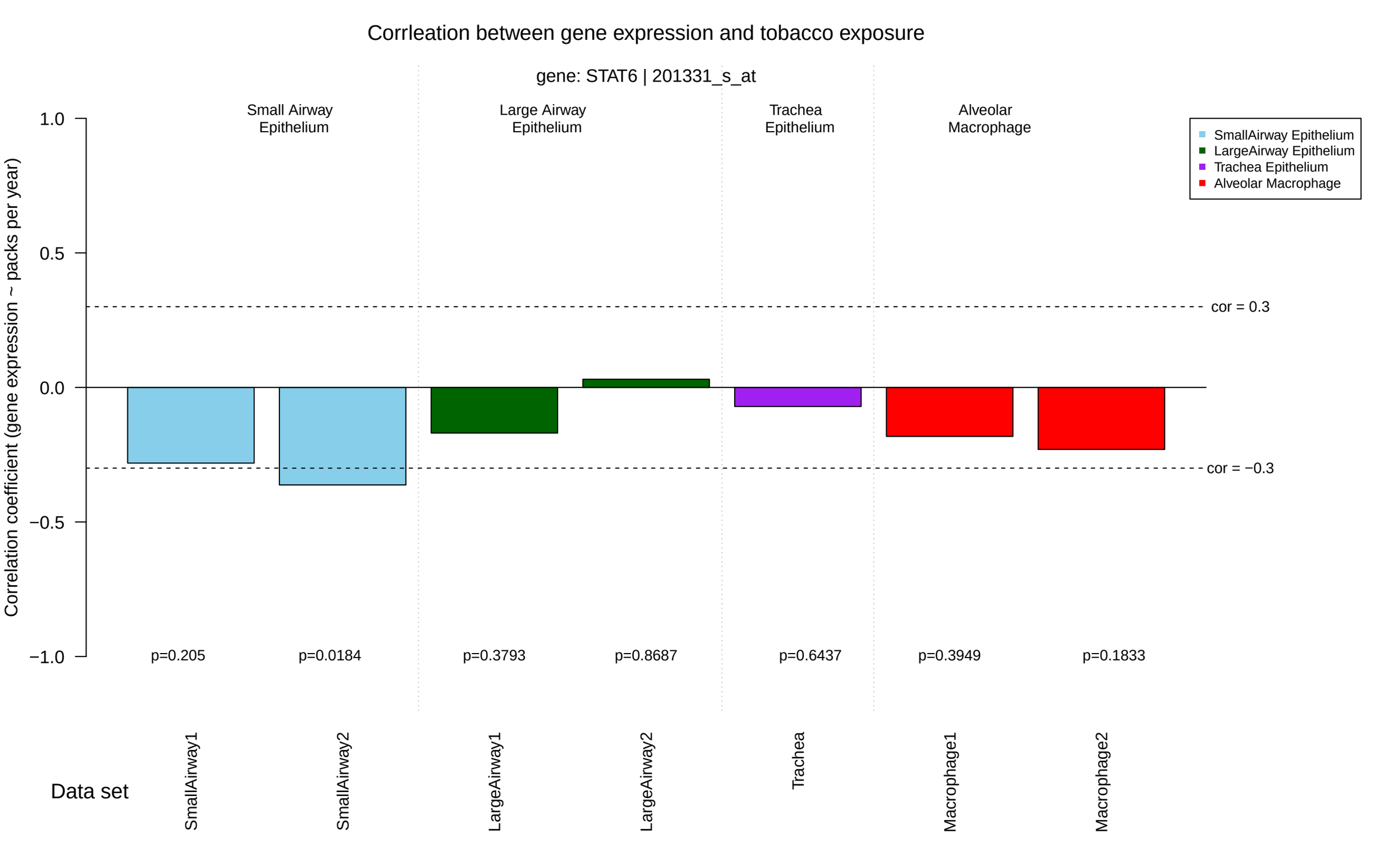

set: LargeAirway1 (p=0.5389)
STAT6 | 201331_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3839)
STAT6 | 201331_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8074)
STAT6 | 201331_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4947)
STAT6 | 201331_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7754)
STAT6 | 201331_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT6 | 201331_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.205
p=0.0184
p=0.3793
p=0.8687
p=0.6437
p=0.3949
p=0.1833
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2